FILTER DESIGN TEST CASE

Lattice Filter - Band 13 (700 MHz) for GPS/GNSS Anti-Jamming Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Lattice Filter filter targeting Band 13 (700 MHz) for GPS/GNSS Anti-Jamming Filter. The filter is designed on 128° YX LiNbO3 substrate with a target center frequency of 4735.9 MHz and bandwidth of 139.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the GPS/GNSS Anti-Jamming Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Lattice Filter
- Frequency Band: Band 13 (700 MHz)
- Application: GPS/GNSS Anti-Jamming Filter
- Substrate: 128° YX LiNbO3
- Package: Fan-Out WLP
- Design Tool: PathWave EM Design
- Design Center: Irvine Design Center

This specification applies to all variants of the Lattice Filter design targeting Band 13 (700 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 4735.9 MHz
Bandwidth (3 dB): 139.1 MHz
Insertion Loss Target: <= 1.3 dB
Return Loss Target: >= 9.0 dB
Out-of-Band Rejection: >= 27.0 dB
Isolation (Duplexer): >= 33.0 dB
Q Factor: 7813
Temperature Coefficient:-6.4 ppm/°C
Die Size: 2.9 x 2.5 mm
Wafer Size: 8-inch
Number of Resonators: 5
Electrode Material: Mo/Al

Passivation: SiO2

4. TEST PROCEDURE

1. Open PathWave EM Design and load the Lattice Filter template project.
2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
3. Set design targets: center freq 4735.9 MHz, BW 139.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Fan-Out WLP).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 128° YX LiNbO3).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 14208 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.3 dB
- Return loss in passband shall be >= 9.0 dB
- Out-of-band rejection at +/- 209 MHz offset >= 27.0 dB
- Group delay variation across passband shall be <= 6 ns
- Temperature drift over -40°C to +85°C shall be <= 3.8 MHz
- Power handling shall be >= +27 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 94%

6. TEST RESULTS

Measurement Date: 2025-04-15
Devices Tested: 54
Insertion Loss (meas): 0.98 dB
Return Loss (meas): 10.8 dB
Rejection (meas): 29.9 dB
Isolation (meas): 37.6 dB
Q Factor (meas): 7813
Group Delay Variation: 5.1 ns
Power Handling: +26 dBm
Die Yield: 88.1%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

a) The Lattice Filter design demonstrated excellent performance for Band 13 (700 MHz).
b) Insertion loss met target specification.
c) Rejection at near-band offset requires additional resonator tuning.
d) Group delay flatness improved compared to previous revision.
e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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Approver:	Dr. Karen Mitchell	Date: 2025-04-28

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